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The Copperbelt University

Kapasa Makasa University Campus

Department of Agriculture and Aquatic Sciences

SESSIONAL EXAMINATIONS 2022

COURSE CODE: NR 430

COURSE NAME: PLANT PROTECTION

TIME ALLOWED: THREE (3) HOURS

DATE: 19 TH OCTOBER 2022

INSTRUCTIONS: 1. Answer the COMPULSORY QUESTION ONE

2. Answer ONE other question from Section A to make it TWO

3. Answer at least ONE question from Section B or C

3. You are required to answer a total of five (5) questions in the exam as follows:

i) 2 from section A including the COMPULSORY question one

ii) At least ONE from BOTH section B and C

iii) You are at liberty to pick ONE OR TWO from section A and Section B

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NR 430 PLANT PROTECTION

SECTION A (Answer Question 1 and ONE other question from this section)

Question One (COMPULSORY)

a) Explain various parameters used in classification and identification of plant diseases.

(12 marks)

b) Describe the various and distinct stages involved in pathogenesis. *(6 marks)*

c) What is dieback? *(2 marks)*

Question Two

a) Discuss disease tetrahedron in detail. *(12 marks)*

b) Define the following terms:

i. Virulence

ii. Phytopathogenesis

iii. Biotrophs

iv. Phytoanticipins

(8 marks)

Question Three

a) Explain the various cultural practices that aim at reducing the adverse effects of plant diseases.

(10 marks)

b) Briefly explain how bacteria reproduce. *(5 marks)*

c) Explain the difference between viruses and viroids. *(5 marks)*

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SECTION B (Answer one or two questions from this section)

Question Four

a) Pests are known to develop resistance to a certain pesticide after a certain period of time. Discuss how this phenomenon occurs. *(5 marks)*

b) Expalin how plants defend themselves against infections using pre-existing defense structures. *(10 marks)*

c) Explain how Enzyme Linked Immuno-sorbent Assay **(ELISA)** works as a molecular method of disease identification. *(5 marks)*

Question Five

a) Describe four (4) major plant viral diseases and how they affect a given plant species. *(10 marks)*

b) Briefly explain how Polymerase Chain Reaction **(PCR)** works as a plant disease identification method. *(5 marks)*

c) List the various symptoms and signs caused by plant bacterial diseases. *(5 marks)*

Section C (Answer one or two questions from this section)

Question Six

a) Discuss the various principles that anchor Integrated Pest Management (IPM). *(8 marks)*

b) Briefly explain various characteristics of arthropods. *(8 marks)*

c) List various cultural practices that can be incorporated into an IPM system. *(4 marks)*

Question Seven

a) Insects are living organisms that belong to the Class *Insecta* and Phylum *Anthropoda*. State and explain five (5) insect "Orders" into which insects can be classified including examples of insect order you have selected. *(10 marks)*

b) Give a detailed explanation of the importance of understanding insect classification as an Agro-forester. *(10 marks)*

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Copperbelt University

KAPASA MAKASA UNIVERSITY CAMPUS

DEPARTMENT OF AGRICULTURAL AND AQUATIC SCIENCES

COURSE: PLANT PROTECTION (NR 430)

2021 SESSIONAL EXAMINATION

Time Allowed: 3hrs **Date:** 9th November 2021

Instructions:

1. This question paper comprises seven (7) questions.
2. Question one (1) is compulsory
3. You are required to attempt a total of five (5) questions as follows: question one (1), any one (1) question of your choice from sections A, B and C, and the fifth question from either section B or C
4. Each question must be properly numbered and answered on a separate sheet of paper

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SECTION A - PATHOLOGY

Question one [Compulsory]

a) Explain with valid reasons why it may be easy or difficult in understanding genetics of plant tolerance and resistance to diseases.

(10 marks)

b) Differentiate *heterokaryosis* from *parasexualism*

(10 marks)

Question two

a) Define *epidemiology* in relation to plants or crops

(5 marks)

b) Explain how the factors causing plant disease interact with one another for the disease to be apparent

(15 marks)

Question three

a) Explain the physical defence mechanisms of plants against diseases

(10 marks)

b) Differentiate induced biochemical changes from induced cellular defence mechanism of plants against diseases

(10 marks)

SECTION B - ENTOMOLOGY

Question four

a) Explain with examples what differentiates one insect order from another

(10 marks)

b) Explain the importance of categorizing major insect orders in Agro-forestry by Agro-foresters.

(10 marks)

Question five

a) The ultimate goal of insect surveillance is to come up with insect' species abundance and richness in a given area. Explain with two (2) examples how this goal is achieved

(10 marks)

b) Insect population dynamics has played a prominent role in the development of basic ecology and in the understanding and management of serious pests over the landscape. Discuss the statement.

(10 marks)

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SECTION C - FIRES

Question six

a) Explain fire behaviour in terms of different types of fuel

b) Explain what a fire triangle is and how it is used by fire fighters (10 marks)

Question seven (10 marks)

a) Define back-firing

b) Describe the three (3) categories of fires (3 marks)

c) Explain the suitability of fire extinguishers in fire fighting (9 marks)

(8 marks)

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The Copperbelt University

Kapasa Makasa University Campus

Department of Agriculture and Aquatic Sciences

SESSIONAL EXAMINATIONS 2023

COURSE CODE: NR 430

COURSE NAME: PLANT PROTECTION

TIME ALLOWED: THREE (3) HOURS

DATE: 23rd OCTOBER 2023

TIME: 09:00hrs

STRUCTIONS:
1. This paper comprises of seven (7) questions
2. Answer the COMPULSORY QUESTION ONE
3. Answer any TWO questions from Section B
4. Answer any TWO questions from Section C
5. You are required to answer a total of five (5) questions in the exam
6. Each question carries 20 marks

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SECTION A

QUESTION 1 (compulsory)

(a) Explain various parameters used in classification and identification of plant diseases

(12 marks)

(b) Explain how Fluorescence In-Situ Hybridization (FISH) works as a molecular method of disease identification. Further state the advantages of this molecular method. **(8 marks)**

Uses fluorescent globe to bind H H NO SH H
Specific DNA Sequence

SECTION B

QUESTION 2

(a) In details, discuss plant disease tetrahedron

(8 marks)

(b) Define the following terms

i. Stridulation

ii. Phytopathogenesis

iii. Biotrophs

iv. Phytoanticipins

(8 marks)

(b) Describe various distinct stages involved in pathogenesis

(4 marks)

QUESTION 3

(a) Insects are living organisms that belong to the class *insecta* and phylum *athropoda*. State and describe five (5) ''*Orders*' into which insects can be classified including examples of insects that belong to such *Orders*. **(10 marks)**

* Hymenoptera – Bees
* Homoptera – Aphids
* Hemiptera – bug
* Lepidoptera – butterfly
* Diptera – Mosquitoes flies
* Orthoptera – Grasshopper lice
* Coleoptera – beetles – 1

(b) Briefly explain how bacteria reproduce **(5 marks)**

(c) Describe the differences between viruses and viroids **(5 marks)**

- Inoculation
- Infection
- en.

chewing mouth
siphoning
chewing sucking
piercing sucking
Respiratory sucking

QUESTION 4

(a) Pests are known to develop resistance to a certain pesticide after a given period of time. Discuss how this phenomenon occur **(5 marks)**

(b) Explain how plants defend themselves against infections using pre-existing defense structures **(10 marks)**

(c) Nematodes are known to cause a number of problems and diseases in plants. Discuss the damages caused by nematodes and how they cause such damages **(5 marks)**

SECTION C

QUESTION 5

(a) Discuss similarities and differences between compound interest diseases and simple interest diseases. **(10 marks)**

(b) Briefly, explain how Polymerase Chain Reaction (PCR) works as a plant disease identification method. * Amplify the specific DNA sequence using primers and enzymes **(5 marks)**

(c) List various symptoms and signs caused by plant bacterial diseases. **(5 marks)**

QUESTION 6

(a) Discuss the four-tiered approach that anchors Integrated Pest Management (IPM). **(8 marks)**

(b) Explain in details how viruses invade the plant and cause diseases **(4 marks)**

(C) Briefly explain various characteristics of arthropods **(8 marks)**

QUESTION 7

(a) Explain various cultural practices that aim at reducing the adverse effects of plant diseases. **(10 marks)**

(b) In details, explain the importance of understanding insect classifications as an Agroforester **(10 marks)**

Specificity

12357

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Question 1

What do you understand by the following terms and give example where necessary.

(i) obligate parasite

(ii) zoospores

(iii) latent virus

(iv) predisposition and pathogenicity

(v) serology.

Question 2

Discuss how viruses are transmitted from the infected plant to health plant. further describe how they move from one cell to another.

Question 3

How do do nematodes attack on the plant leads to the over branching of roots while in ~~some roots white~~ in some species results in hyperplasia.

KAPASA MAKASA UNIVERSITY

DEPARTMENT OF AGRICULTURAL AND AQUATIC SCIENCES

ASSESSMENT 3

COURSE CODE: NR450, SA 330

TIME ALLOWED: 1hr: 30 min

DATE: 25TH SEPTEMBER, 2024

INSTRUCTIONS

- ✓ This question paper comprises of THREE (3) questions
- ✓ You are required to ANSWER ALL the questions
- ✓ Allocation of marks is indicated at the end of each question.

- Disease symptoms
- Type of organ
- affected plant type
- Type of phytopathogen.

1. (a) Explain various parameters used in classification and identification of plant diseases **(8 marks)**

- Inoculation
- Pre penetration
- penetration
- colonization
- infection

(b) Describe various distinct stages involved in plant disease cycle **(18 marks)**

(invasion) colonization
- Growth
- Reproduction

2. (a) Pests are known to develop resistance to a certain pesticide after a certain period of time. Discuss how this phenomenon occur **(5 marks)**

action threshhold

Identify and monitor pest.

Prevention

control

3. Discuss the four-tiered approach that crop producers ought to follow in Integrated Pest Management **(12 marks)**

4. Clearly, distinguish between polyclic and monocyclic diseases **(7 marks)**

- Spread
- dissemination seeds, soils
- Produced vector
- generation
- incubation

Total Marks = 50

PLANT Protection TEST ONE 2023

. What do you understand by the following term?

(a) - Hemibiotroph (2)

b) - Disease syndrome (2)

c) - Necrotroph (2)

d) - Predisposition (2)

State and explain different parameters used to identify and describe plant disease 5. (12)

Describe various symptoms associated with necrosis in Fungal disease Infestation in plants (10)